7.2.7 Theorem *If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, then $f \in \mathcal{R}[a, b]$.*

Proof. It follows from Theorem 5.4.3 that f is uniformly continuous on $[a, b]$. Therefore, given $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $u, v \in [a, b]$ and $|u - v| < \delta_\varepsilon$, then we have $|f(u) - f(v)| < \varepsilon/(b - a)$.

Let $\mathcal{P} = \{I_i\}_{i=1}^n$ be a partition such that $||\mathcal{P}|| < \delta_\varepsilon$. Applying Theorem 5.3.4 we let $u_i \in I_i$ be a point where f attains its minimum value on I_i, and let $v_i \in I_i$ be a point where f attains its maximum value on I_i.

Let α_ε be the step function defined by $\alpha_\varepsilon(x) := f(u_i)$ for $x \in [x_{i-1}, x_i)$ $(i = 1, \ldots, n - 1)$ and $\alpha_\varepsilon(x) := f(u_n)$ for $x \in [x_{n-1}, x_n]$. Let ω_ε be defined similarly using the points v_i instead of the u_i. Then one has

$$\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) \qquad \text{for all} \qquad x \in [a, b].$$

Moreover, it is clear that

$$\begin{aligned} 0 \le \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) &= \sum_{i=1}^n (f(v_i) - f(u_i))(x_i - x_{i-1}) \\ &< \sum_{i=1}^n \left(\frac{\varepsilon}{b - a}\right)(x_i - x_{i-1}) = \varepsilon. \end{aligned}$$

Therefore it follows from the Squeeze Theorem that $f \in \mathcal{R}[a, b]$. Q.E.D.

Monotone functions are not necessarily continuous at every point, but they are also Riemann integrable.

7.2.8 Theorem *If $f : [a, b] \to \mathbb{R}$ is monotone on $[a, b]$, then $f \in \mathcal{R}[a, b]$.*

Proof. Assume that f is increasing on $I = [a, b]$. Partitioning the interval into n equal subintervals $I_k = [x_{k-1}, x_k]$ gives us $x_k - x_{k-1} = (b - a)/n$, $k = 1, 2, \ldots, n$. Since f is increasing on I_k, its minimum value is attained at the left endpoint x_{k-1} and its maximum value is attained at the right endpoint x_k. Therefore, we define the step functions $\alpha(x) := f(x_{k-1})$ and $\omega(x) := f(x_k)$ for $x \in [x_{k-1}, x_k)$, $k = 1, 2, \ldots, n - 1$, and $\alpha(x) := f(x_{n-1})$ and $\omega(x) := f(x_n)$ for $x \in [x_{n-1}, x_n]$. Then we have $\alpha(x) \le f(x) \le \omega(x)$ for all $x \in I$, and

$$\int_a^b \alpha = \frac{b - a}{n}(f(x_0) + f(x_1) + \cdots + f(x_{n-1}))$$

$$\int_a^b \omega = \frac{b - a}{n}(f(x_1) + \cdots + f(x_{n-1}) + f(x_n)).$$

Subtracting, and noting the many cancellations, we obtain

$$\int_a^b (\omega - \alpha) = \frac{b - a}{n}(f(x_n) - f(x_0)) = \frac{b - a}{n}(f(b) - f(a)).$$

Thus for a given $\varepsilon > 0$, we choose n such that $n > (b - a)(f(b) - f(a))/\varepsilon$. Then we have $\int_a^b (\omega - \alpha) < \varepsilon$ and the Squeeze Theorem implies that f is integrable on I. Q.E.D.

The Additivity Theorem

We now return to arbitrary Riemann integrable functions. Our next result shows that the integral is an "additive function" of the interval over which the function is integrated. This property is no surprise, but its proof is a bit delicate and may be omitted on a first reading.

7.2.9 Additivity Theorem *Let $f := [a,b] \to \mathbb{R}$ and let $c \in (a,b)$. Then $f \in \mathcal{R}[a,b]$ if and only if its restrictions to $[a, c]$ and $[c, b]$ are both Riemann integrable. In this case*

$$\int_a^b f = \int_a^c f + \int_c^b f. \tag{6}$$

Proof. ($\Leftarrow$) Suppose that the restriction f_1 of f to $[a, c]$, and the restriction f_2 of f to $[c, b]$ are Riemann integrable to L_1 and L_2 respectively. Then, given $\varepsilon > 0$ there exists $\delta' > 0$ such that if $\dot{\mathcal{P}}_1$ is a tagged partition of $[a, c]$ with $||\dot{\mathcal{P}}_1|| < \delta'$, then $\left|S(f_1; \dot{\mathcal{P}}_1) - L_1\right| < \varepsilon/3$. Also there exists $\delta'' > 0$ such that if $\dot{\mathcal{P}}_2$ is a tagged partition of $[c, b]$ wilh $||\dot{\mathcal{P}}_2|| < \delta''$ then $\left|S(f_2; \dot{\mathcal{P}}_2) - L_2\right| < \varepsilon/3$. If M is a bound for $|f|$, we define $\delta_\varepsilon := \min\{\delta', \delta'', \varepsilon/6M\}$ and let $\dot{\mathcal{P}}$ be a tagged partition of $[a, b]$ with $||\dot{\mathcal{Q}}|| < \delta$. We will prove that

$$\left|S(f;\dot{\mathcal{Q}}) - (L_1 + L_2)\right| < \varepsilon. \tag{7}$$

(i) If c is a partition point of $\dot{\mathcal{Q}}$, we split $\dot{\mathcal{Q}}$ into a partition $\dot{\mathcal{Q}}_1$ of $[a, c]$ and a partition $\dot{\mathcal{Q}}_2$ of $[c, b]$. Since $S(f;\dot{\mathcal{Q}}) = S(f;\dot{\mathcal{Q}}_1) + S(f;\dot{\mathcal{Q}}_2)$, and since $\dot{\mathcal{Q}}_1$ has norm $< \delta'$ and $\dot{\mathcal{Q}}_2$ has norm $< \delta''$, the inequality (7) is clear.

(ii) If c is not a partition point in $\dot{\mathcal{Q}} = \{(I_k, t_k)\}_{k=1}^m$, there exists $k \le m$ such that $c \in (x_{k-1}, x_k)$. We let $\dot{\mathcal{Q}}_1$ be the tagged partition of $[a, c]$ defined by

$$\dot{\mathcal{Q}}_1 := \{(I_1, t_1), \ldots, (I_{k-1}, t_{k-1}), ([x_{k-1}, c], c)\},$$

and $\dot{\mathcal{Q}}_2$ be the tagged partition of $[c, b]$ defined by

$$\dot{\mathcal{Q}}_2 := \{([c, x_k], c), (I_{k+1}, t_{k+1}), \ldots, (I_m, t_m)\}.$$

A straightforward calculation shows that

$$\begin{aligned} S(f;\dot{\mathcal{Q}}) - S(f;\dot{\mathcal{Q}}_1) - S(f;\dot{\mathcal{Q}}_2) &= f(t_k)(x_k - x_{k-1}) - f(c)(x_k - x_{k-1}) \\ &= (f(t_k) - f(c)) \cdot (x_k - x_{k-1}), \end{aligned}$$

whence it follows that

$$\left|S(f;\dot{\mathcal{Q}}) - S(f;\dot{\mathcal{Q}}_1) - S(f;\dot{\mathcal{Q}}_2)\right| \le 2M(x_k - x_{k-1}) < \varepsilon/3.$$

But since $||\dot{\mathcal{Q}}_1|| < \delta \le \delta'$ and $||\dot{\mathcal{Q}}_2|| < \delta \le \delta''$, it follows that

$$\left|S(f;\dot{\mathcal{Q}}_1) - L_1\right| < \varepsilon/3 \qquad \text{and} \qquad \left|S(f;\dot{\mathcal{Q}}_2) - L_2\right| < \varepsilon/3,$$

from which we obtain (7). Since $\varepsilon > 0$ is arbitrary, we infer that $f \in \mathcal{R}[a,b]$ and that (6) holds.

($\Rightarrow$) We suppose that $f \in \mathcal{R}[a,b]$ and, given $\varepsilon > 0$, we let $\eta_\varepsilon > 0$ satisfy the Cauchy Criterion 7.2.1. Let f_1 be the restriction of f to $[a, c]$ and let $\dot{\mathcal{P}}_1$, $\dot{\mathcal{Q}}_1$ be tagged partitions of

$[a, c]$ with $||\dot{\mathcal{P}}_1|| < \eta_\varepsilon$ and $||\dot{\mathcal{Q}}_1|| < \eta_\varepsilon$. By adding additional partition points and tags from $[c, b]$, we can extend $\dot{\mathcal{P}}_1$ and $\dot{\mathcal{Q}}_1$ to tagged partitions $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ of $[a, b]$ that satisfy $||\dot{\mathcal{P}}|| < \eta_\varepsilon$ and $||\dot{\mathcal{Q}}|| < \eta_\varepsilon$. If we use the *same* additional points and tags in $[c, b]$ for both $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$, then

$$S(f_1; \dot{\mathcal{P}}_1) - S(f_1; \dot{\mathcal{Q}}_1) = S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}}).$$

Since both $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ have norm η_ε, then $|S(f_1; \dot{\mathcal{P}}_1) - S(f_1; \dot{\mathcal{Q}}_1)| < \varepsilon$. Therefore the Cauchy Condition shows that the restriction f_1 of f to $[a, c]$ is in $\mathcal{R}[a, c]$. In the same way, we see that the restriction f_2 of f to $[c, b]$ is in $\mathcal{R}[c, d]$.

The equality (6) now follows from the first part of the theorem. Q.E.D.

7.2.10 Corollary *If $f \in \mathcal{R}[a, b]$, and if $[c, d] \subseteq [a, b]$, then the restriction of f to $[c, d]$ is in $\mathcal{R}[c, d]$.*

Proof. Since $f \in \mathcal{R}[a, b]$ and $c \in [a, b]$, it follows from the theorem that its restriction to $[c, b]$ is in $\mathcal{R}[c, b]$. But if $d \in [c, b]$, then another application of the theorem shows that the restriction of f to $[c, d]$ is in $\mathcal{R}[c, d]$. Q.E.D.

7.2.11 Corollary *If $f \in \mathcal{R}[a, b]$ and if $a = c_0 < c_1 < \cdots < c_m = b$, then the restrictions of f to each of the subintervals $[c_{i-1}, c_i]$ are Riemann integrable and*

$$\int_a^b f = \sum_{i=1}^{m} \int_{c_{i-1}}^{c_i} f.$$

Until now, we have considered the Riemann integral over an interval $[a, b]$ where $a < b$. It is convenient to have the integral defined more generally.

7.2.12 Definition If $f \in \mathcal{R}[a, b]$ and if $\alpha, \beta \in [a, b]$ with $\alpha < \beta$, we define

$$\int_\beta^\alpha f := -\int_\alpha^\beta f \qquad \text{and} \qquad \int_\alpha^\alpha f := 0.$$

7.2.13 Theorem *If $f \in \mathcal{R}[a, b]$ and if α, β, γ are any numbers in $[a, b]$, then*

$$\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f, \tag{8}$$

in the sense that the existence of any two of these integrals implies the existence of the third integral and the equality (8).

Proof. If any two of the numbers α, β, γ are equal, then (8) holds. Thus we may suppose that all three of these numbers are distinct.

For the sake of symmetry, we introduce the expression

$$L(\alpha, \beta, \gamma) := \int_\alpha^\beta f + \int_\beta^\gamma f + \int_\gamma^\alpha f.$$

It is clear that (8) holds if and only if $L(\alpha, \beta, \gamma) = 0$. Therefore, to establish the assertion, we need to show that $L = 0$ for all six permutations of the arguments α, β, and γ.

We note that the Additivity Theorem 7.2.9 implies that $L(\alpha, \beta, \gamma) = 0$ when $\alpha < \gamma < \beta$. But it is easily seen that both $L(\beta, \gamma, \alpha)$ and $L(\gamma, \alpha, \beta)$ equal $L(\alpha, \beta, \gamma)$.

Moreover, the numbers

$$L(\beta, \alpha, \gamma), \qquad L(\alpha, \gamma, \beta), \qquad \text{and} \qquad L(\gamma, \beta, \alpha)$$

are all equal to $-L(\alpha, \beta, \gamma)$. Therefore, L vanishes for all possible configurations of these three points. Q.E.D.

Exercises for Section 7.2

1. Let $f : [a, b] \to \mathbb{R}$. Show that $f \notin \mathcal{R}[a, b]$ if and only if there exists $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}$ there exist tagged partitions $\dot{\mathcal{P}}_n$ and $\dot{\mathcal{Q}}_n$ with $||\dot{\mathcal{P}}_n|| < 1/n$ and $||\dot{\mathcal{Q}}_n|| < 1/n$ such that $\left|S(f; \dot{\mathcal{P}}_n) - S(f; \dot{\mathcal{Q}}_n)\right| \geq \varepsilon_0$.
2. Consider the function h defined by $h(x) := x + 1$ for $x \in [0, 1]$ rational, and $h(x) := 0$ for $x \in [0, 1]$ irrational. Show that h is not Riemann integrable.
3. Let $H(x) := k$ for $x = 1/k$ ($k \in \mathbb{N}$) and $H(x) := 0$ elsewhere on [0, 1]. Use Exercise 1, or the argument in 7.2.2(b), to show that H is not Riemann integrable.
4. If $\alpha(x) := -x$ and $\omega(x) := x$ and if $\alpha(x) \leq f(x) \leq \omega(x)$ for all $x \in [0, 1]$, does it follow from the Squeeze Theorem 7.2.3 that $f \in \mathcal{R}[0, 1]$?
5. If J is any subinterval of $[a, b]$ and if $\varphi_J(x) := 1$ for $x \in J$ and $\varphi_J(x) := 0$ elsewhere on $[a, b]$, we say that φ_J is an *elementary step function* on $[a, b]$. Show that every step function is a linear combination of elementary step functions.
6. If $\psi : [a, b] \to \mathbb{R}$ takes on only a finite number of distinct values, is ψ a step function?
7. If $S(f; \dot{\mathcal{P}})$ is any Riemann sum of $f : [a, b] \to \mathbb{R}$, show that there exists a step function $\varphi : [a, b] \to \mathbb{R}$ such that $\int_a^b \varphi = S(f; \dot{\mathcal{P}})$.
8. Suppose that f is continuous on $[a, b]$, that $f(x) \geq 0$ for all $x \in [a, b]$ and that $\int_a^b f = 0$. Prove that $f(x) = 0$ for all $x \in [a, b]$.
9. Show that the continuity hypothesis in the preceding exercise cannot be dropped.
10. If f and g are continuous on $[a, b]$ and if $\int_a^b f = \int_a^b g$, prove that there exists $c \in [a, b]$ such that $f(c) = g(c)$.
11. If f is bounded by M on $[a, b]$ and if the restriction of f to every interval $[c, b]$ where $c \in (a, b)$ is Riemann integrable, show that $f \in \mathcal{R}[a, b]$ and that $\int_c^b f \to \int_a^b f$ as $c \to a+$. [*Hint:* Let $\alpha_c(x) := -M$ and $\omega_c(x) := M$ for $x \in [a, c)$ and $\alpha_c(x) := \omega_c(x) := f(x)$ for $x \in [c, b]$. Apply the Squeeze Theorem 7.2.3 for c sufficiently near a.]
12. Show that $g(x) := \sin(1/x)$ for $x \in (0, 1]$ and $g(0) := 0$ belongs to $\mathcal{R}[0, 1]$.
13. Give an example of a function $f : [a, b] \to \mathbb{R}$ that is in $\mathcal{R}[c, b]$ for every $c \in (a, b)$ but which is not in $\mathcal{R}[a, b]$.
14. Suppose that $f : [a, b] \to \mathbb{R}$, that $a = c_0 < c_1 < \cdots < c_m = b$ and that the restrictions of f to $[c_{i-1}, c_i]$ belong to $\mathcal{R}[c_{i-1}, c_i]$ for $i = 1, \ldots, m$. Prove that $f \in \mathcal{R}[a, b]$ and that the formula in Corollary 7.2.11 holds.
15. If f is bounded and there is a finite set E such that f is continuous at every point of $[a, b] \backslash E$, show that $f \in \mathcal{R}[a, b]$.
16. If f is continuous on $[a, b]$, $a < b$, show that there exists $c \in [a, b]$ such that we have $\int_a^b f = f(c)(b - a)$. This result is sometimes called the *Mean Value Theorem for Integrals*.
17. If f and g are continuous on $[a, b]$ and $g(x) > 0$ for all $x \in [a, b]$, show that there exists $c \in [a, b]$ such that $\int_a^b fg = f(c) \int_a^b g$. Show that this conclusion fails if we do not have $g(x) > 0$. (Note that this result is an extension of the preceding exercise.)